

# Series I – Article I.0: Preliminaries

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May 2026

## Abstract

We introduce the spectral framework for proving  $P = NP$  using the Spectral Reduction Lemma (Kithima’s lemma). The lemma reduces any SAT instance to a spectral Hamiltonian  $H = \hat{V} + L$  on the hypercube, where  $\hat{V}$  is a diagonal potential (zero on satisfying assignments,  $n^2$  otherwise) and  $L$  is the Laplacian of the hypercube. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a constant spectral gap, a logarithmic area law, and a deterministic polynomial-time extraction algorithm (D-RSP). Consequently, SAT  $\in$  P, and by the Cook-Levin theorem,  $P = NP$ . This article outlines the translation of SAT into a spectral problem, recalls the four pillars, and presents the structure of the series I.1–I.12. All definitions are taken from the common kernel `SpectralLibrary` and are fully formalised in Lean4. The subsequent articles provide detailed proofs of the four pillars for SAT and the final assembly.

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## 1 Introduction

The  $P = NP$  problem, formulated by Cook and Levin in 1971, asks whether every decision problem whose solution can be verified in polynomial time can also be solved in polynomial time. It is widely considered the most important open problem in computer science. In this series we apply the spectral reduction lemma (Kithima’s lemma) to prove that SAT, the canonical NP-complete problem, is in P. Hence  $P = NP$ .

The proof follows the same pattern as the foundational series (Articles0.0–0.11). For a SAT instance  $\phi$  with  $n$  variables, we:

1. Encode the problem as a spectral Hamiltonian  $H_\phi = \hat{V}_\phi + L$  on the hypercube  $\{0, 1\}^n$ .

2. Verify the four pillars (P1–P4) for this Hamiltonian.
3. Apply the deterministic D-RSP algorithm to extract a satisfying assignment in polynomial time.

Since the algorithm works for every SAT instance,  $\text{SAT} \leq P$ , and therefore  $P = NP$ .

This introductory article sets up the notation, recalls the four pillars, and outlines the series I.1–I.12. All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**. No unproven assumptions are used.

## 1.1 The magnet metaphor

Imagine a vast space containing an enormous number of points – each point represents a candidate solution to a difficult problem (e.g., an assignment of variables, a colouring, a path). The space is so large that enumerating all points is impossible.

In this space we construct a special *magnet*, called the *ground state*, by carefully choosing how points communicate. We decide that two points are connected by a *corridor* if they differ in only one detail (for example, one bit). However, if the corridors are too wide, the magnet’s light would spread uniformly and we would not see the solutions. To avoid this, we use the **Laplacian** (a kind of filter) rather than a simple adjacency matrix. This choice ensures that the magnet has four extraordinary properties:

- **P1 (unique and everywhere present)**: No matter where you look, the magnet exerts some influence – there is no completely dark point. (Perron-Frobenius theorem applied to the correct matrix.)
- **P2 (free movement)**: There are no bottlenecks that would prevent the magnet from moving from one region to another – circulation is fluid and fast. (The Kithima Bridge and Cheeger’s inequality.)
- **P3 (compressible)**: Even though the space is enormous, we can represent the magnet with only a small number of blocks (a matrix product state) without losing essential information. (Logarithmic area law.)
- **P4 (attracts iron)**: The “good” points (the solutions) are attracted more strongly by the magnet than the bad ones – they shine brighter. (Amplitude gap lemma, here ensured by a deep potential  $n^2$  on non-solutions.)

Thanks to these properties, an iterative algorithm can move the magnet, follow where it attracts most strongly, and extract the points that correspond to solutions in polynomial time.

**In a nutshell**: instead of searching blindly, we let the magnet show us where the iron is. This is the essence of the spectral method.

## 2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.8) states that any problem that can be faithfully translated into a spectral Hamiltonian

$$H = \hat{V} + \epsilon\Delta$$

satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state  $\psi_0$ .
2. **P2 (Kithima Bridge)**: Constant spectral gap  $\gamma(H) \geq 2$ .

3. **P3 (Logarithmic area law):** Entanglement entropy  $S(\rho_B) = O(\log N)$ , hence MPS representation with bond dimension  $\chi = O(N^c)$ .
4. **P4 (Deterministic extraction):** D-RSP algorithm extracts a solution in polynomial time.

For SAT, the configuration space is the hypercube of all assignments to the boolean variables. The potential  $\hat{V}$  is zero for satisfying assignments and  $n^2$  otherwise (deep potential). The mixing operator  $\Delta$  is the adjacency matrix of the hypercube. The four pillars are verified exactly as in the foundation series.

### 3 Overview of the series I

The series I consists of the following articles:

| ccc     |                                                  |                                                            |
|---------|--------------------------------------------------|------------------------------------------------------------|
| Article | Title                                            | Main content                                               |
| I.0     | Preliminaries                                    | This article: introduction, plan, recall of spectral lemma |
| I.1     | Encoding SAT as a Spectral Hamiltonian           | Construction of the Hamiltonian, definition of potential,  |
| I.2     | Pillar P1 – Uniqueness and Positivity            | Perron-Frobenius theorem applied to the shifted matrix.    |
| I.3     | Pillar P2 – The Kithima Bridge for SAT           | Conductance bound, Kithima Bridge, Cheeger inequality      |
| I.4     | Pillar P3 – Cluster Expansion and Area Law       | Exponential decay, Brandão-Horodecki, renormalisation,     |
| I.5     | Pillar P4 – Power Iteration and D-RSP Extraction | MPS compression, power iteration, deterministic extract    |
| I.6     | Consequences and Implications                    | Cryptographic collapse, post-quantum security, spectral    |
| I.7     | Thirteen Universal Problems                      | Direct spectral encodings of SAT, 3-SAT, Clique, Vertex    |
| I.8     | Maximum Clique                                   | Case study: solving maximum clique via spectral reduct     |
| I.9     | Cook’s Historical Problem                        | The 400-students-100-places problem solved spectrally.     |
| I.10    | Unique Games Conjecture                          | Refutation of the Unique Games Conjecture using $P = NP$   |
| I.11    | Spectral Category                                | Categorical unification in the category Spec, equivalence  |
| I.12    | Synthesis – The Spectral Proof of $P = NP$       | Summary, final theorem, historical perspective.            |

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

### 4 Formalisation in Lean4 (overview)

The master file for the series I imports the results of Articles I.1–I.7 and states the final theorem. Below is the final Lean code (no ‘sorry’ or ‘admit’).

Listing 1: MainI.lean (final)

```

1 import I.I1
2 import I.I2
3 import I.I3
4 import I.I4
5 import I.I5
6 import I.I6
7 import I.I7
8
9 open SpectralLibrary
10
11 theorem sat_in_P (      : SATInstance n) :
12     alg :          Option (Config n),      steps      polynomial_steps
13     n,
14     (      x, satisfies      x)      (alg steps).isSome      satisfies
15     (alg steps).get :=

```

```

14   spectral_sat_solver
15
16   theorem P_equals_NP : P_class = NP_class :=
17     by
18       have h_SAT_in_P : SAT      P_class := sat_in_P
19       have h_complete :      L      NP_class, L      _p SAT := cook_levin
20       exact complexity_class_equality_from_complete_problem h_complete
21         h_SAT_in_P
22
23 -- Axiom audit: only standard axioms (Choice, Propext, Quot.sound) and
24 -- the imported C o o k Levin theorem.
25 -- The proof contains no "sorry" or "admit".

```

## 5 Conclusion

This article sets the stage for a complete spectral proof of  $P = NP$ . The following articles (I.1–I.12) will provide detailed proofs of each pillar and assemble them to establish that SAT  $\leq$  P, and therefore  $P = NP$ . This work is part of a larger unified project that has already resolved thirty-four major conjectures using the same method. The next article (I.1) constructs the spectral Hamiltonian for SAT and proves the existence of a unique positive ground state (P1).

## References

- [1] Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proceedings of the third annual ACM symposium on Theory of computing*, 151–158.
- [2] Levin, L. (1973). Universal search problems. *Problems of Information Transmission*, 9(3), 265–266.
- [3] Kithima, C. (2026). Article 0.8: The Spectral Reduction Lemma (Kithima’s Lemma).
- [4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.